

Whose HLA is in your reference?

Five projects. One graph of the MHC, a highly variable immune region.

754

MHC haplotypes

462 Asian / Arab haplotype entries
266 East Asian + 72 South Asian + 124 Arab
752 donor haplotypes + 2 references
376 source sample entries; 371 distinct donors.

Samples by source



HPRC: 232 worldwide; JaSaPaGe: 19 total

MHC structural backbone: 1,677 segments, 2,381 links

1000 Genomes ($n = 2504$)

THE PAYOFF

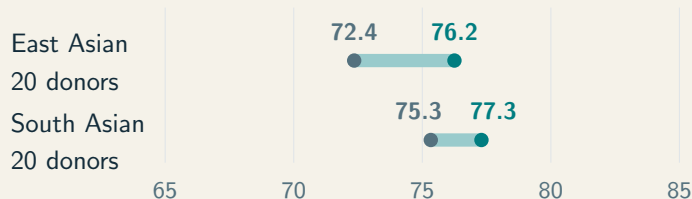
More haplotypes. Better SV calls.

79,844 SNV sites and 298 SV-containing graph sites per donor.

SV-bearing genotypes recovered exactly (%)

● HPRC-only panel

● Expanded panel



+3.9

points in East Asian donors

95% CI: +2.4 to +5.3

South Asian: +2.0 points

95% CI: +0.3 to +3.5

SV-bearing truth genotypes: 1,465 EAS + 1,577 SAS.

Shared-SNV comparison: 1,007,636 genotype comparisons per ancestry group.

40-donor pilot; assembly truth. Test families excluded from panels; test assemblies remain in graph topology.

THE TYPER

DogoHLA

A pangenome-native, population-specific HLA typer.

Collect reads

AsianPGR haplotypes



Call and phase

Read-backed variants



Reconstruct

Graph-supported indels



Type HLA

Sequences + HLA names

Uses SpecHLA components for alignment and small variants; panel graphs supply supported noncoding changes.

Preliminary: exact whole-gene sequences

8 development donors × 8 genes × 2 haplotypes

Native SpecHLA



36/128

DogoHLA



57/128

AsianPGR class II / DRB1 structure



Reference illustration; colour = haplotype sharing.

22.8% fewer sequence edits; correct two-field genotypes: 61/63 → 62/63.

Selected development results; further independent analysis is needed.

Global whole-gene matches to assembly truth; larger independent evaluation ongoing.